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Fundamentals of IT & AI

Module 1: The Software Application Life Cycle

- What is an Application?
- Types of Applications
- Web Application Fundamentals
- Web Technologies:
  - **Frontend:** HTML, CSS, JavaScript, React
  - **Backend:** Python, Java, Node.js
  - **Databases:** SQL (MySQL, PostgreSQL), NoSQL (MongoDB)
- Software Development Life Cycle (SDLC)
  - **Phases:** Planning, Analysis, Design, Implementation (Coding), Testing, Deployment, Maintenance.
- Application Development Methodologies
  - **Agile:** Core principles, Scrum, Kanban
  - **Waterfall**

Module 2: Data Fundamentals

- What is Data?
- Types of Data
- Data Storage
- Data Analysis
- Data Engineering
- Data Science

Module 3: Computing and the Cloud

- The Importance of Computing Power
- Key Computing Technologies:
  - CPU (Central Processing Unit)
  - GPU (Graphics Processing Unit)
- Cloud Computing:
  - What is the Cloud?
  - Cloud Service Models:
    - IaaS (Infrastructure as a Service)
    - PaaS (Platform as a Service)
    - SaaS (Software as a Service)

Module 4: Introduction to AI and Generative AI

- What is Artificial Intelligence (AI)?
- How AI Works?
- Key Concepts:
  - Machine Learning (ML)
  - Deep Learning (DL)
- Generative AI:
  - What is Generative AI?
  - Examples: Large Language Models (LLMs), image generation models.
- AI in Everyday Learning

Module 5: Real-World Applications of Technology

- Customer Relationship Management (CRM)
- Human Resource Management Systems (HRMS)
- Retail & E-Commerce

- Healthcare

Basic Python

Module 1: Python Introduction and Setup

- Python's applicability across various domains
- Installation, environment setup, and path configuration
- Writing and executing the first Python script

Module 2: Python Fundamentals

- Keywords, Identifiers, and basic syntax
- Variables, Data Types, and Operators
- Introduction to Input/Output operations

Module 3: Control Structures and Functions

- Conditional Statements: If, Elif, Else
- Loops: For, While, and control flow mechanisms
- Understanding and defining Functions in Python

Module 4: Strings and Collections

- String operations and manipulations
- Lists and their operations
- Introduction to Tuples and Sets

Module 5: Diving Deeper into Collections

- Detailed exploration of Dictionaries
- Frozen Sets and their use-cases
- Advanced list comprehensions

Advanced Python

Module 1: Advanced Collections and Sequences

- Advanced methods in Lists, Tuples, and Dictionaries
- Sets, Frozen Sets, and operations
- Comprehensive look into Python Collections

Module 2: Functional Programming in Python

- Exploring types of Functions and Arguments
- Lambda functions and their applications
- Map, Reduce, and Filter functions

Module 3: File Handling and Modules

- File operations and handling different file formats
- Working with Excel and CSV files in Python
- Understanding and using Python Modules and Packages

Module 4: Object-Oriented Programming

- Deep dive into Classes, Objects, and Methods
- Constructors, Destructors, and Types of Methods
- Inheritance, Polymorphism, and Encapsulation

Module 5: Exception Handling and Regular Expressions

- Exception Handling: Try, Except, Finally
- Creating and using Custom Exceptions
- Utilizing Regular Expressions for pattern matching

Tools

AI-powered coding assistant for efficiency and automation.

Git is a distributed version control system for tracking code changes.

Jenkins is an open-source tool for automating CI/CD pipelines.

Kubernetes automates deployment, scaling, and management of containers.

AI model for generating and assisting with code and text.

Google's AI model for assisting in code generation, debugging, and research.

Power BI is a business analytics tool for visualizing data insights.

Version control tools for managing and collaborating on test automation scripts.

Python powers everything from web apps to AI.

Core framework for building autonomous AI agents and workflows.

Tools

Linux is an open-source operating system for servers, desktops, and devices.

Nexus is a repository manager for managing build artifacts and dependencies.

Slack is a messaging platform for teams and workplace collaboration.

Trello is a visual tool for organizing tasks and projects.

SQL is a language for managing and querying relational databases.

Lightweight and customizable code editor with Python support.

Interactive coding environment for testing DSA concepts.

One of the best platforms for solving DSA problems.

Step-by-step visualization of code execution.